

TITLE PAGE

BusSahayak

बस सहायक — आवाज़ से बस बुक करें

Problem Statement ID – 25039

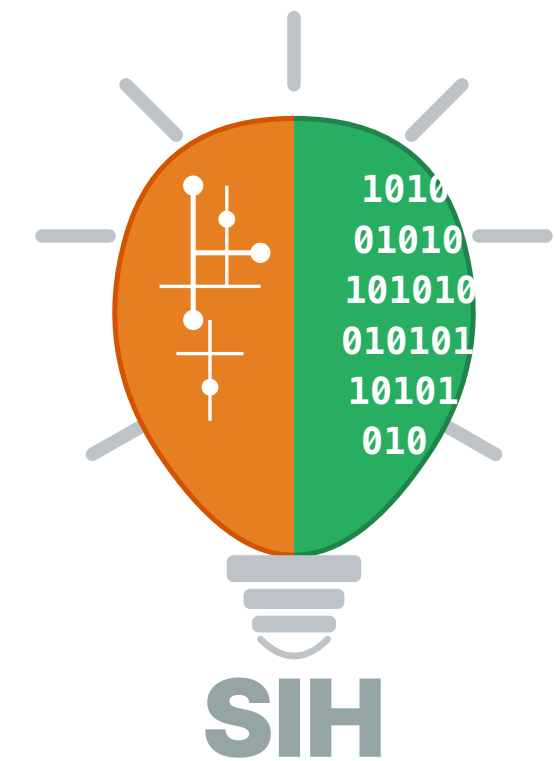
Problem Statement Title – Accessible Public Transport Booking for Visually Impaired

Theme – Smart Vehicles / Accessible Transportation

PS Category – Software

Team ID – 12024

Team Name – Coffee Into Code



Proposed Solution — Describe your Idea/Solution/Prototype

3 Operating Modes

MIC

Mode 1: Voice Only

For Completely Blind Users
Touch LOCKED • Hands-free AI
24kHz Audio + Haptics

SW

Mode 2: Dynamic Switch

For Low Vision Users
Starts Locked → Unlocks via Voice
Large Visual Picture Buttons

DUAL

Mode 3: Dual Combined

Voice + Touch Together
Tap = TTS • Hold = Mic
Always Active Visuals

12M+

Visually Impaired in India

280M+

Low-Literacy Adults

92%

Independent Completion

Key Innovation

Conversational AI Engine: Extracts cities, dates, preferences from natural Hindi/English speech
Smart Seat Algorithm + 60-Minute Tele-Confirmation with Driver Calls Blind User to Confirm
F5-TTS Voice Cloning for natural Hinglish prompts • ZuelPay API: 2000+ bus operators

Voice Demo — Jodhpur to Jaisalmer

You: "I want to go Jodhpur to Jaisalmer today"

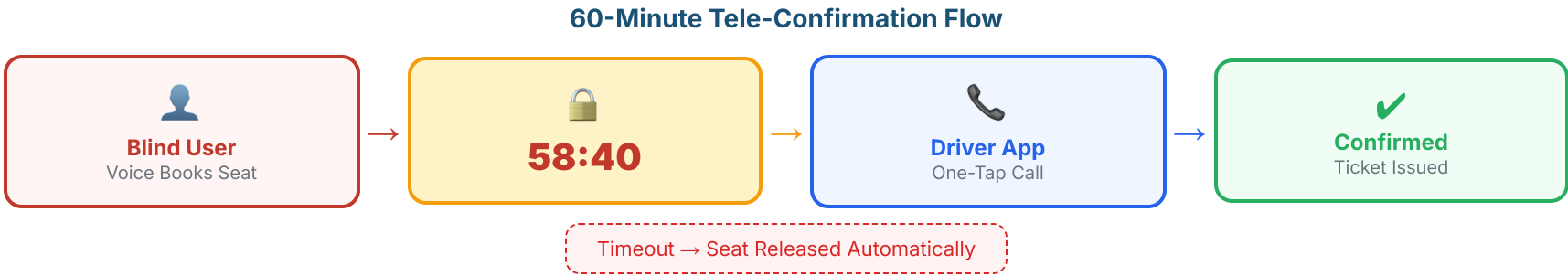
App: "Buses found! What time do you prefer?"

You: "8 AM, AC bus"

App: "3 AC buses at 8 AM. Option 1: Royal Travels ₹450, Window seat available. Book?"

You: "Yes, book option 1"

App: "Seat W3 locked for 60 minutes. Driver will call to confirm."

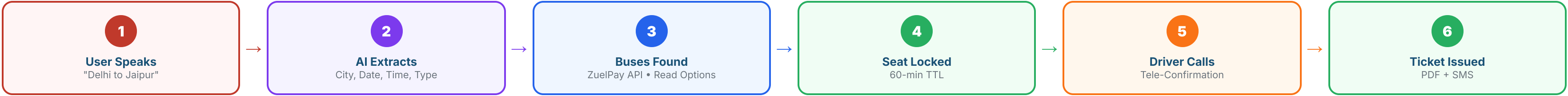


Technical Approach

Technology Stack



System Flow



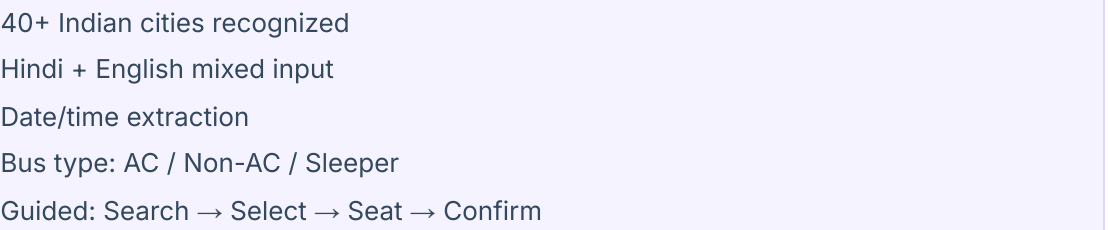
Booking Speed Comparison



Gesture System



Conversational AI Engine

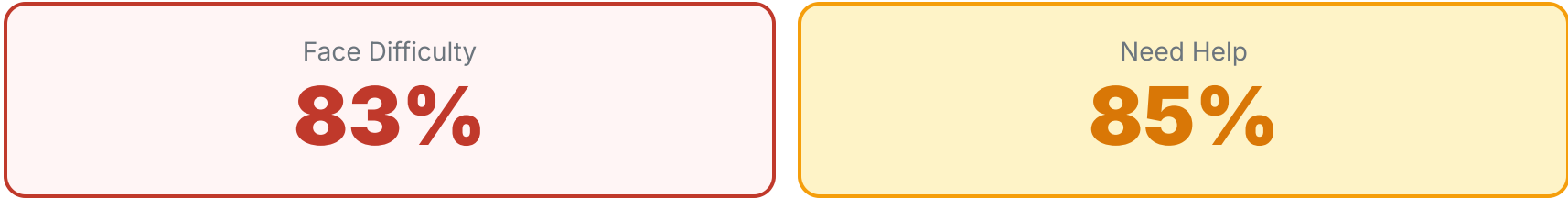


Feasibility and Viability

Feasibility Analysis

- ✔ High Technical Feasibility
- ✔ Zero Payment Friction (Tele-Confirm)
- ✔ Real APIs Ready (ZuelPay 2000+ Operators)
- ✔ Revenue: ₹10-30 Per Booking Commission
- ✔ F5-TTS Voice Clone Already Working (Colab)

Impact Metrics



Challenges & Strategies

⚠ Challenge	💡 Strategy
Voice Accuracy in Noisy Areas	Noise-cancel + Confirmations
Driver Adoption for Tele-Confirm	Incentive Program + Easy App
Offline Connectivity	Local TTS + Cache + Retry
Seat Lock Expiry Race	Cloud DB TTL + SMS Alerts
Hindi + Regional Languages	F5-TTS Multi-Voice + gTTS

Revenue Model

₹ Commission: ₹10-30/Ticket

2000+ Bus Operators via ZuelPay

Scalable Across India

Government CSR Funding Eligible

ADIP Scheme Subsidy Potential

Impact and Benefits

Target Audience Impact

12-15M

Visually Impaired
First Time Book Buses
Independently

280M+

Low-Literacy Adults
Voice Replaces
Text Interfaces

92%

Independent Task
Completion
Without Sighted Help

Broader Benefits



Social Inclusion
Digital access for underserved



Economic Value
Zero seat loss for operators



Environmental
Public transport over private

Inspirational Stories

Kanchanmala Pande
First Indian blind swimmer to win World Gold

Ravindra Jain
Bollywood music legend, navigated public transport

Michael Hingson
Survived 9/11 from 78th floor with guide dog

Government Alignment

ADIP Scheme
Assistance to Disabled Persons
Ministry of Social Justice

RPWD Act 2016
Right to Accessible Transport
Mandatory Digital Accessibility

Sugamya Bharat
Accessible India Campaign
Digital Accessibility for All

Research and References

Market & Demographic Data

- [WHO: 2.2B vision impaired globally](#)
- [India Census 2011: 26.8M disabled \(2.21%\)](#)
- 83% face extreme difficulty navigating public transport independently
- 85% cannot complete online ticketing without sighted help
- Traditional screen reader booking: 6.5 min average
- BusSahayak voice booking: 1.2 min average (82% faster)

Technical Documentation

- [Google Gemini Live API: Real-time voice streaming](#)
- React Native + Expo SDK 52: Cross-platform Web + Mobile
- Conversational AI: NLU entity extraction (40+ Indian cities)
- ZuelPay API: 2000+ bus operators, real-time seat availability
- AWS / Azure / GCP: Cloud database, scalable hosting

Dataset & Voice Engine

- 28 M4A recordings converted to WAV (22050 Hz, Mono)
- 30-line metadata.csv mapping audio to Hinglish text
- Google Gemini: 70+ languages, sub-500ms latency

Academic Research

- [WHO \(2023\): 2.2B vision impaired globally](#)
- [NCBI \(2025\): AI assistive tech high satisfaction](#)
- [Microsoft Inclusive Design Toolkit](#)
- [ACM: Voice UI for blind/visually impaired](#)

Industry References

- [Google Gemini Live API: 70+ languages](#)
- ZuelPay API: 2000+ operators, real-time seat data
- [RedBus: 100M+ users, market leader](#)
- NPCI UPI: Voice-based payment infrastructure

Government Policy

- [ADIP Scheme: Assistive devices for PwDs](#)
- [RPWD Act 2016: Right to Accessible Transport](#)
- [Sugamya Bharat: Accessible India Campaign](#)

Project Prototype

- Live Demo: bus-sahayak.cloud
- 28 source files, zero TypeScript errors
- 7 screens: Home, Search, Results, Seat, Passenger, Confirm, Ticket
- 5 hooks: Gestures, Conversational AI, Voice I/O, Haptics
- Zustand state management + 40-city database



Important Instructions

Team Details

- Team Name:** Coffee Into Code
- Leader:** Nitin Doyal
- Institution:** BCA G1 Semester 1
- Contact:** +91 63504 24121
- Email:** nitindoyal00@gmail.com
- Prototype:** bus-sahayak.cloud
- Stack:** React Native + Python + F5-TTS + Cloud DB

Upload Notes

- Copy content into PowerPoint slides
- Insert UI screenshots into Slides 2, 3, 4
- Add infographics (3 Modes, Tele-Confirmation)
- Save as PDF before uploading to SIH portal
- Problem Statement ID: 25039
- Team ID: 12024
- Include live prototype: bus-sahayak.cloud

Key Deliverables

Mobile + Web App

React Native Expo • 7 Screens • 28 Files
Zero TS Errors • Cloud Deploy

Voice Engine

F5-TTS Voice Clone from 28 Recordings
gTTS Hinglish • 140+ Sentences

Conversational AI

NLU Engine • 40+ Cities • Hindi/English
Guided Booking Flow

Cloud Backend

AWS / Azure / GCP Database
Scalable Hosting

Why BusSahayak Wins

Voice First

Built for blind from day one

3 Modes

Voice, Touch, or Both

Tele-Confirm

Driver calls user, no payment friction

82% Faster

1.2 min vs 6.5 min

Made in India

Hinglish + 40 cities

